

Randomized Comparison of Cisplatin Plus Fluorouracil and Carboplatin Plus Fluorouracil Versus Methotrexate in Advanced Squamous-Cell Carcinoma of the Head and Neck: A Southwest Oncology Group Study

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Purpose: The Southwest Oncology Group (SWOG) conducted a randomized comparison of cisplatin plus fluorouracil (5-FU) and carboplatin plus 5-FU versus single-agent methotrexate (MTX) in patients with recurrent and metastatic squamous-cell carcinoma (SCC) of the head and neck. The primary objective was to compare separately the response rates of each combination regimen to standard weekly MTX.

Patients and Methods: Two hundred seventy-seven patients diagnosed with SCC of the head and neck were randomized to one of three treatments: (1) cisplatin 100 mg/m² intravenously (IV) on day 1 and 5-FU 1,000 mg/m² per day for a 96-hour continuous infusion repeated every 21 days; (2) carboplatin 300 mg/m² IV on day 1 and 5-FU 1,000 mg/m² per day for a 96-hour continuous infusion repeated every 28 days; and (3) MTX 40 mg/m² IV given weekly.

Results: All three treatment regimens were well tolerated. However, both hematologic and nonhematologic toxicities were significantly greater with cisplatin plus 5-FU compared with MTX ($P = .001$). Toxicity from carbo-

platin plus 5-FU was intermediate compared with the other regimens. The complete and partial response rates were 32% for cisplatin plus 5-FU, 21% for carboplatin plus 5-FU, and 10% for MTX. The comparison of cisplatin plus 5-FU to MTX was statistically significant ($P < .001$), and the comparison of carboplatin plus 5-FU to MTX was of borderline statistical significance ($P = .05$). Median response durations and median survival times were similar for all three treatment groups. Logistic regression models showed that only treatment assigned was associated significantly with response ($P = .001$). Cox proportional hazards models indicated that only performance status was associated significantly with survival ($P < .01$).

Conclusions: We conclude that combination chemotherapy resulted in improved response rates but was associated with an increased toxicity and no improvement in overall survival. Therefore, new treatments that may alter the course of disease in recurrent head and neck cancer patients still need to be identified.

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THE MOST COMMONLY given anticancer agent that is used worldwide for the palliation of patients with recurrent squamous-cell carcinoma (SCC) of the head and neck is probably methotrexate (MTX). Response rates that used conventional dosing schedules were reported to vary between 8% and 50% and averaged approximately 30%.¹ The advantages of MTX are its ease of administration in an outpatient setting and the low incidence of serious side effects. MTX is considered the standard treatment of this disease to which new agents or combinations are compared.

Cisplatin had a similar response rate reported from phase II trials that used doses of 80 to 120 mg/m² every 3 to 4 weeks.² In two trials that directly compared MTX and cisplatin in recurrent disease patients, no differences were observed in the response rates or the duration of response, but toxicity was greater with cisplatin.^{3,4}

In an effort to improve response rates and survival, numerous two- and three-drug combination regimens have been tested in recurrent head and neck cancer patients.^{5,6} The drugs frequently studied in combination include bleomycin, MTX, cisplatin, vinblastine or vincris-

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tine, and fluorouracil (5-FU).⁶ Although the results of single-institution pilot trials often have suggested benefit, no improvements in survival have been observed in prospective randomized trials that compared combinations to single-agent cisplatin or MTX.⁷⁻¹¹ Additionally, toxicities of several of these agents overlapping significantly increased toxicity.

Cisplatin that is given in combination with any drug that is excreted through the kidneys has the potential to cause life-threatening toxicity should renal tubular damage occur and delay drug excretion. Nausea, vomiting, ototoxicity, and peripheral neuropathy are other side effects that are either potentially dose-limiting or limit patient eligibility and compliance. Despite these limitations, one cisplatin-based combination that seems to be well tolerated and efficacious in both recurrent disease and previously untreated patients is cisplatin and infusional 5-FU.^{12,13}

Carboplatin was selected for clinical development because of its more favorable toxicity profile. The dose-limiting toxicity was myelosuppression, whereas nephrotoxicity, ototoxicity, and neurotoxicity rarely were observed. The Southwest Oncology Group (SWOG) evaluated carboplatin in a phase II trial in recurrent head and neck cancer and observed a 24% response rate.¹⁴ A logical next step was the evaluation of carboplatin and 5-FU in combination. Forastiere et al¹⁵ recommended a starting dose of 300 mg/m² of carboplatin in combination with a 5-day infusion of 5-FU based on the results of a phase II trial.

These studies led the SWOG to design a phase III trial in 1985 to determine, in large numbers of patients, the antitumor activity and toxicity of cisplatin plus 5-FU and carboplatin plus 5-FU and to compare both regimens with MTX.

PATIENTS AND METHODS

The major objective of this three-arm study was to compare separately the response rates of cisplatin plus 5-FU and carboplatin plus 5-FU with standard-treatment MTX in patients with locally recurrent or metastatic head and neck cancer. Eligible patients had histologically proven SCC of the head and neck that was either recurrent after attempted cure with surgery and radiation therapy (RT) or newly diagnosed disease with distant metastases. Those patients with recurrent disease had received no prior chemotherapy for treatment of the recurrence, although they could have received induction chemotherapy 6 or more months before study entry. All patients were required to have measurable disease, a life expectancy of at least 12 weeks, and a performance status of 0, 1, or 2 on the SWOG criteria. A 24-hour creatinine clearance $\geq$ 50 mg/mL, a WBC count greater than 3,500, a platelet count greater than 100,000, and a normal serum calcium were required. All patients gave informed consent in accordance with Food and Drug Administration and institutional guidelines.

Patients were randomized to one of three treatments: (1) cisplatin 100 mg/m² intravenously (IV) on day 1 and 5-FU 1,000 mg/m² every 24 hours for a 96-hour continuous infusion that was repeated every 21 days; (2) carboplatin 300 mg/m² IV on day 1 and 5-FU 1,000 mg/m² every 24 hours for a 96-hour continuous infusion that was repeated every 28 days; and (3) MTX 40 mg/m² IV per week. Doses were modified based on the degree of myelosuppression (nadir counts of the previous cycle), stomatitis, or dermatitis; in addition, cisplatin was modified for the occurrence of renal toxicity, ototoxicity, and neurotoxicity. There was no dose escalation for cisplatin, but the dose was reduced to 75 mg/m² for the occurrence of grade 3 myelosuppression and to 60 mg/m² for grade 4 toxicity. There was one dose escalation for carboplatin to 360 mg/m² for grade 0 or 1 myelosuppression and a dose reduction to 240 mg/m² for grade 3 or 4 toxicity. MTX was increased to 50 mg/m² weekly for grade 0 mucositis or myelosuppression and was reduced by 10 mg/m² for grade 2 or greater toxicity. The 5-FU dosage was not modified for myelosuppression. The daily 5-FU dose was reduced to 800 mg/m² for grade 2 or more mucosal or skin toxicity.

Cisplatin was mixed with normal saline and was administered during a 15- to 30-minute period with pretreatment and posttreatment hydration and mannitol diuresis. Carboplatin was administered during a 5- to 10-minute period with no hydration. Standard antiemetic regimens were used by the individual investigators.

At the time of registration and randomization, patients were stratified by the following categories: (1) performance status 0-1 versus 2; (2) prior induction chemotherapy with cisplatin; (3) prior RT; (4) newly diagnosed metastatic disease versus recurrent disease. Standard criteria were used to assess response. A CR was defined as the disappearance of all clinical evidence of tumor for a minimum of 4 weeks. Partial response (PR) was defined as a 50% or more decrease in the sum of the products of all diameters of measured lesions for a minimum of 4 weeks without the appearance of any new lesions. Stable disease (SD) was defined as a steady state or a decrease in measurable lesions less than would qualify for a PR without the worsening of symptoms or the appearance of new lesions for a minimum of 4 weeks. Progression was defined as the unequivocal increase of at least 25% in the size of any measurable lesion, appearance of new lesions, or uncontrolled hypercalcemia. Response duration was measured as the time from which CR or PR was first documented until the first documentation of relapse or the last disease assessment date. Survival was measured from the time of randomization to death. Toxicity was reported as the worst grade (SWOG criteria) experienced while on treatment.

χ^2 tests were used to compare the response and toxicity differences between the treatment arms.¹⁶ Because of the large number of toxicity comparisons made, only those with a *P* value less than .01 were interpreted as being statistically significant. Median response duration, median survival time, and the survival curves were estimated by using the Kaplan-Meier method.¹⁷ Logistic regression models were used to test for the association of assigned treatment and pretreatment patient characteristics with response.¹⁶ Treatment differences in the survival curves were tested by using log-rank tests.¹⁸ Proportional hazards models were used to test for the association of treatment and pretreatment patient characteristics with survival.¹⁹ All reported *P* values were two-sided. Analyses were based on all eligible patients unless otherwise stated.

RESULTS

Between February 1986 and March 1989, 277 patients were registered onto the study. Sixteen patients were deemed ineligible for the following reasons: no measurable disease (six patients), newly diagnosed locally advanced disease with no metastatic involvement (five patients), history of prior lung cancer (two patients), and one each for elevated creatinine clearance, radiotherapy within 2 weeks of registration, and prior chemotherapy for recurrent disease. Patient pretreatment characteristics are given in Table 1. The treatment arms were balanced for all characteristics. Seventy-two percent of patients had a performance status of 0 or 1, 7% were newly diagnosed with distant metastases, and 11% had received prior cisplatin as part of an induction chemotherapy regimen.

Treatment

Of the 261 eligible patients, two did not start treatment. One patient randomized to cisplatin plus 5-FU died from pneumonia before treatment began, and the other, although randomized to MTX, elected to have noncurative surgery after randomization and never started chemotherapy.

Table 1. Patient Characteristics

	Cisplatin Plus 5-FU (N = 87)		Carboplatin Plus 5-FU (N = 86)		MTX (N = 88)	
	No.	%	No.	%	No.	%
Age, years						
Median	61.0		61.0		60.0	
Range	39-82		43-77		28-80	
Sex						
Males	76	87	71	83	73	83
Females	11	13	15	17	15	17
Race						
White	67	77	71	83	68	77
Black	17	20	14	16	15	17
Other	3	3	1	1	5	6
Performance status						
0-1	63	72	61	71	63	72
2	24	28	25	29	25	28
Disease						
Advanced*	6	7	4	5	8	9
Recurrent	81	93	82	95	80	91
Prior chemotherapy with cisplatin†						
Yes	10	11	9	10	10	11
No	77	89	77	90	78	89
Prior RT						
Yes	78	90	75	87	79	90
No	9	10	11	13	9	10

*Newly diagnosed with distant metastases.

†Induction chemotherapy only at least 6 months before study entry.

Table 2. Distribution of Courses Received by Each Treatment Arm

No. of Courses	Cisplatin Plus 5-FU		MTX*		Carboplatin Plus 5-FU		MTX†	
	No.	%	No.	%	No.	%	No.	%
0-1	18	21	17	19	22	26	26	30
2	16	18	22	25	24	28	25	28
3-4	27	31	25	29	22	26	24	27
5 or more	26	30	24	27	18	20	13	15
Total	87	100	88	100	86	100	88	100

*Three weeks of MTX was equal to one course of cisplatin plus 5-FU.

†Four weeks of MTX was equal to one course of carboplatin plus 5-FU.

The number of courses of chemotherapy by treatment arm is listed in Table 2. Because cisplatin plus 5-FU was given at 3-week intervals and carboplatin plus 5-FU was given at 4-week intervals, separate comparisons were made to MTX. Three weeks of MTX were considered equal to one course of cisplatin plus 5-FU and 4 weeks of MTX equal to one course of carboplatin plus 5-FU. The median number of courses received by patients on cisplatin plus 5-FU was four (range, zero to 17). For carboplatin plus 5-FU, the median was two (range, one to 14), and the median number of weeks on MTX was 8 (range, 0 to 49).

MTX should have been escalated if there were no toxicities in the preceding week. Among the 85 patients who received more than 1 week of MTX, 24 (28%) should have had their dose escalated in the first month of treatment and did not. Carboplatin should have been escalated if grade 0 or 1 myelosuppression occurred in the preceding cycle. Among the 64 patients who received more than one course of carboplatin plus 5-FU, 21 (33%) should have had their dose escalated and did not.

Toxicity

Three patients were not assessable for toxicity; two did not start treatment and one died early from nontreatment-related causes before toxicities could be documented. Among the 258 patients who were assessable for toxicity, there were three treatment-related deaths, one on each arm. All three had grade 4 hematologic toxicity that led to sepsis. Hematologic toxicity is detailed in Table 3. Grades 3 and 4 granulocytopenia occurred infrequently in all three arms. Leukopenia was most common in the cisplatin-treated patients; 26 (33%) had a nadir WBC count of less than 2,000 μ L. Thrombocytopenia was more frequent in carboplatin-treated patients. The overall maximum grade of hematologic toxicity (categorized as none v grades 1 to 2 v grades 3+) was significantly worse for cisplatin plus 5-FU treatment compared with MTX ($P < .001$). The difference be-

Table 3. Hematologic Toxicity

Toxicity	Cisplatin Plus 5-FU (N = 85)					Carboplatin Plus 5-FU (N = 86)					MTX (N = 87)				
	Grade					Grade					Grade				
	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4
Anemia	34	13	34	4	0	38	11	25	10	2	62	10	12	3	0
Granulocytopenia	51	10	17	3	4	66	8	10	1	1	74	6	1	1	5
Leukopenia	23	12	24	24	2	33	17	26	8	2	48	12	13	9	5
Thrombocytopenia	69	4	7	2	3	59	10	6	5	6	74	3	5	2	3
Maximum grade any toxicity No.	9	10	38	23	5	20	14	30	15	7	38	11	24	8	6

tween carboplatin plus 5-FU and MTX was also suggestive of a difference ($P = .02$).

Nonhematologic toxicities are listed in Table 4. Stomatitis, nausea, and vomiting were the most frequently reported side effects. Renal toxicity was more common on the cisplatin arm and was least common on the carboplatin arm. Comparisons of any toxicity versus no renal toxicity were highly significant for cisplatin plus 5-FU versus MTX ($P < .001$), and were not significant for carboplatin plus 5-FU versus MTX. One anaphylactic reaction occurred on the second day of 5-FU administration in a cisplatin plus 5-FU treatment patient. Six cardiac events were observed with cisplatin plus 5-FU: hypotension, tachycardia, chest pain, myocardial infarction with hypotension, and atrial fibrillation. One cardiac event (hypotension) was reported with MTX; none were reported with carboplatin plus 5-FU.

χ^2 tests for differences in the distribution of worst overall toxicity (hematologic and nonhematologic categorized as none ν grades 1 to 2 ν grades 3+) indicated that patients who were treated with cisplatin plus 5-FU experienced significantly more toxicity than those who were treated with MTX ($P < .001$). The comparison of carboplatin plus 5-FU to MTX was nonsignificant. The exclusion of nausea and vomiting from the comparisons gave similar results.

Table 4. Nonhematologic Toxicity

Toxicity	Cisplatin Plus 5-FU (N = 85)		Carboplatin Plus 5-FU (N = 86)		MTX (N = 87)	
	Grade (%)		Grade (%)		Grade (%)	
	1-2	3-4	1-2	3-4	1-2	3-4
Diarrhea	12	2	6	2	3	0
Stomatitis	19	14	28	15	34	10
Nausea/vomiting	68	8	48	6	38	8
Peripheral neuropathy	5	1	2	0	0	0
Ototoxicity	8	4	2	0	2	0
Renal	18	9	1	1	3	3

Response and Survival

All eligible patients (261) were included in the analyses for response and survival. Response data are listed in Table 5 and survival curves are depicted in Fig 1. The overall response rate (CR and PR) was 32% for cisplatin plus 5-FU, 21% for carboplatin plus 5-FU, and 10% for MTX. The comparison of cisplatin plus 5-FU to MTX is statistically significant ($P < .001$). The comparison of carboplatin plus 5-FU with MTX is of borderline statistical significance ($P = .05$). The median response duration is 4.2 months for cisplatin plus 5-FU, 5.1 months for carboplatin plus 5-FU, and 4.1 months for MTX. Nine CRs were observed: five on cisplatin plus 5-FU (durations, 5.3, 13.3, 20.7, 36+, and 50.1+ months), two on carboplatin plus 5-FU (durations, 3.7 and 7.3 months), and two on MTX (durations, 7.1 and 37.0+ months). All CRs had a performance status of 0 or 1 and recurrent disease.

Logistic regression models were used to test if patient characteristics other than the treatment arm were related significantly to response, and if the control of these other variables altered the unadjusted response comparisons. The models included all three arms. The variables analyzed were age ($< 60 \nu \geq 60$), performance status (0 to 1 ν 2), disease (recurrent ν newly diagnosed with metastases), prior RT, prior cisplatin induction chemotherapy, sex, race, and treatment. Only the treatment

Table 5. Response

	Cisplatin Plus 5-FU		Carboplatin Plus 5-FU		MTX	
	Grade (%)		Grade (%)		Grade (%)	
	No.	%	No.	%	No.	%
Complete response	5	6	2	2	2	2
Partial response	23	26	16	19	7	8
Stable/no response	32	37	36	42	44	50
Increasing disease	14	16	22	25	28	32
Assumed no response*	13	15	10	12	7	8
Total	87	100	86	100	88	100

*Early death or not assessable.

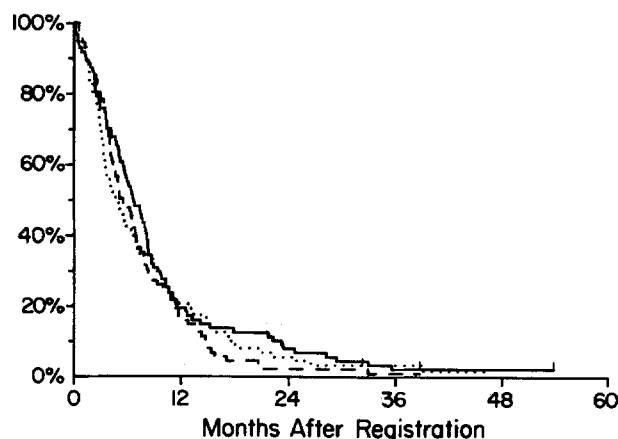


Fig 1. Overall survival by treatment arm of eligible patients with follow-up. —, cisplatin plus 5-FU, 87 patients at risk, 85 deaths, median survival time (MST) 6.6 months; ---, carboplatin plus 5-FU, 86 patients at risk, 85 deaths, MST 5.0 months; —·—, MTX, 88 patients at risk, 87 deaths, MST 5.6 months.

assigned was associated significantly with response ($P = .001$).

The survival of the 261 patients in the three treatment arms is shown in Fig 1. Three patients with a CR (two cisplatin plus 5-FU; one, MTX) and one with a PR (carboplatin plus 5-FU) are alive at a follow-up of 2.7, 3.2, 3.2, and 4.5 years. The median survival times are 6.6 months for cisplatin plus 5-FU, 5.0 months for carboplatin plus 5-FU, and 5.6 months for MTX. There is no significant difference in survival times among the three treatment arms. The percentage of patients who survived at least 9 months was similar for all three treatments: 31% for cisplatin plus 5-FU, 30% for carboplatin plus 5-FU, and 27% for MTX.

Cox proportional hazards models that used the same eight variables previously mentioned to evaluate response associations indicated that only performance status was associated significantly with survival ($P < .01$). The hazards ratio that compared cisplatin plus 5-FU to MTX adjusting for performance status was .88 (95% confidence interval [CI], .65 to 1.20), and compared carboplatin plus 5-FU with MTX was .97 (95% CI, .72 to 1.31).

DISCUSSION

This phase III, prospective randomized trial demonstrated a significant improvement in response rate for the combination cisplatin plus 5-FU compared with weekly MTX. However, this was achieved with significantly more toxicity and without an improvement in overall survival.

In 1985, Kish et al¹² reported a 70% overall response rate with 27% CRs in 30 patients with recurrent and disseminated SCC of the head and neck treated with cisplatin and infusional 5-FU. Since that time, at least a dozen single-institution trials that tested this regimen in recurrent-disease patients have been published.^{6,20} Reported response rates vary between 11% and 79%, with an average of 30 patients accrued to each study.

Cisplatin plus 5-FU has probably become the most widely used regimen for treating head and neck cancer in the United States today. Only two large randomized trials (present study and Jacobs et al²¹) that compared cisplatin plus 5-FU by using the dose and schedule originally published by Kish et al¹² with single agents have been reported. In our study, which used weekly MTX as the control arm, a 32% response rate was observed in 87 patients with recurrent and metastatic disease. Jacobs et al²¹ compared cisplatin plus 5-FU with single-agent cisplatin and single-agent 5-FU and also observed a 32% response rate in 79 patients treated with the combination. This was significantly better than either single agent; the response to single-agent cisplatin was 17%, and the response to single-agent 5-FU 13%. The median survival time was similar to ours, 5.7 months for all patients. Thus, the results of these two large, multiinstitutional trials have served to establish an expected response rate of approximately 32% in recurrent-disease patients treated with cisplatin and infusional 5-FU.

Authors of single-institution pilot trials of carboplatin and infusional 5-FU in recurrent head and neck cancer patients report responses in 32% to 48%.^{15,22} The 21% response rate observed in our randomized trial was low and not different from the results reported for single-agent carboplatin in phase II trials that used 400 mg/m² every 4 weeks.^{14,23,24} In retrospect, our dosing schedule was conservative. Although myelosuppression was the dose-limiting toxicity for carboplatin, the cisplatin plus 5-FU arm had a higher incidence of grade 3 and 4 toxicity. Dose modification parameters were aimed at the achievement of grade 2 myelosuppression, but many investigators failed to escalate carboplatin as dictated by the protocol.

The difference in the median number of courses between the cisplatin plus 5-FU and carboplatin plus 5-FU arms (four v two courses) was not surprising. The cisplatin arm had more responders and fewer patients with progressive disease than carboplatin. The responders would have stayed on treatment longer, whereas most progressive patients would have been taken off treatment by the third course. The difference is also

partly the result of the timing; cisplatin was given every 3 weeks, and carboplatin was given every 4 weeks. By 8 weeks, a cisplatin patient would have received three courses, whereas a carboplatin patient would just be coming in for their third course.

This trial was not designed for a direct statistical comparison between cisplatin plus 5-FU and carboplatin plus 5-FU, but rather as two phase III trials with a shared control arm. A three-arm study with sufficient power to detect a small difference between the two platinum-containing arms and to make comparisons to MTX was not considered feasible. A much larger sample size and a longer period of accrual would have been required. Additionally, the cycle lengths differed, 3 weeks for cisplatin plus 5-FU and 4 weeks for carboplatin plus 5-FU. Because 400 mg/m² of carboplatin is considered equivalent to 100 mg/m² of cisplatin based on pharmacokinetic data, this resulted in an unequal dose intensity for both platinum and 5-FU components. However, the goal was to compare separately the two platinum plus 5-FU regimens to MTX and cisplatin plus 5-FU as piloted by Wayne State University investigators,¹² and carboplatin plus 5-FU by using the recommended dose from phase I trials.¹⁵

The results for carboplatin plus 5-FU in this trial compared with MTX approached statistical significance ($P = .05$); however, more aggressive dose escalation might have demonstrated a clear difference. Considering nonhematologic toxicities, carboplatin plus 5-FU was well tolerated and could be administered in the outpatient setting with the use of infusion ports and ambulatory pumps.

It is of interest that six cardiac events were reported in patients who received cisplatin plus 5-FU, but none were observed in the carboplatin plus 5-FU patients. There is a growing literature of cardiotoxicity reported in association with infusional 5-FU. Studies in humans

suggest three possible mechanisms. Ischemic changes have been observed in up to 68% of patients who underwent cardiac monitoring during 5-FU infusion and have occurred primarily in patients with underlying coronary artery disease.²⁵ Renal losses of magnesium, calcium, and potassium caused by cisplatin are hypothesized to contribute to cardiotoxicity by precipitating arrhythmias.²⁶ 5-FU infusion has been associated with an increase in fibrinopeptide A levels and a decrease in tissue plasminogen activator that may promote thrombosis.^{27,28} We can only speculate that the cardiotoxicity observed in our patients was related to these reported effects of cisplatin and 5-FU on the cardiovascular, metabolic, and coagulation systems.

The treatment of recurrent-disease patients with drugs that have established efficacy in SCC of the head and neck provides limited palliation. In the majority of patients who respond, transient diminution of pain and size of the tumor can be expected. Occasionally, CR occurs and is associated with a long survival, as was observed in three of 261 patients who were treated in this trial. Patients with a good performance status should be encouraged to enroll in investigational protocols that evaluate new agents and biomodulators of cisplatin and 5-FU, whereas for the patient with poor performance and large tumor bulk, supportive care may be a more appropriate plan of action.

We conclude that cisplatin plus 5-FU in the dose and schedule administered in this trial achieved a significantly higher overall response rate when compared with weekly MTX. However, because survival was not different, it is clear that new treatments need to be investigated in this patient population. These results underscore the need to enroll patients in investigational protocols to identify new effective anticancer agents and to compare them with established treatments.

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